SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I K.Niharika(192411124) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

SOLUTIONS

Question 1: Doctor-Shift Assignment Using Backtracking Search

Problem: Assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that all the given constraints are satisfied.

1.1 Constraints

1. D1 cannot work the Night shift.
2. D2 must work before D3: Morning < Afternoon < Night.
3. Only one doctor can be assigned to each shift.
4. D3 cannot work the Morning shift.
5. All doctors must be assigned exactly one shift.

1.2 State Representation

State: (D1, D2, D3), where each value is a shift.

Initial state: (Unassigned, Unassigned, Unassigned)

Goal state: A complete assignment that satisfies all constraints.

1.3 Backtracking Search - Intermediate Steps

Backtracking assigns one doctor at a time. When a choice violates a constraint, that choice is rejected and the algorithm goes back to the previous step.

Step	Choice	Constraint Check	Decision
1	D1 = Morning	D1 is not Night; Morning is free	Accept
2	D2 = Morning	Morning is already occupied by D1	Reject and backtrack
3	D2 = Afternoon	Afternoon is free	Accept
4	D3 = Morning	D3 cannot work Morning	Reject and backtrack
5	D3 = Night	Night is free; Afternoon comes before Night	Accept
6	All assigned	Exactly one doctor per shift and all constraints satisfied	Goal reached

1.4 Constraint Verification

Constraint	Verification	Result
D1 cannot work Night	D1 = Morning	Satisfied
D2 must work before D3	D2 = Afternoon, D3 = Night	Satisfied
Only one doctor per shift	Each shift has exactly one doctor	Satisfied
D3 cannot work Morning	D3 = Night	Satisfied
All doctors assigned once	D1, D2 and D3 each occur once	Satisfied

1.5 Final Valid Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Final Answer: D1 -> Morning, D2 -> Afternoon, D3 -> Night.

Question 2 : Robot Grid Search Using Manhattan Distance

Problem: A robot operates in a 5 x 5 grid. It must move from Start (S) to Goal (G). Obstacles block some paths. Movement is allowed only Up, Down, Left and Right, and every move costs 1.

2.1 Important Algorithm Note

The question mentions BFS, Manhattan Distance and a priority queue together. Standard BFS uses a FIFO queue and does not use a heuristic. A priority queue with the Manhattan heuristic corresponds to A* Search. Therefore, the heuristic-based procedure below is written as A* search.

BFS rule: FIFO queue, no heuristic.

A* rule: Priority queue ordered by f(n) = g(n) + h(n).

2.2 Manhattan Distance

Heuristic: h(n) = |row(n) - row(G)| + |column(n) - column(G)|

Path cost: g(n) = number of moves from S to n, because every move has cost 1.

Evaluation: f(n) = g(n) + h(n)

2.3 Step-by-Step Search Procedure

1. Place S in the priority queue with g(S) = 0.
2. Remove the node with the smallest f(n).
3. Expand the current node and generate Up, Down, Left and Right neighbours.
4. Reject any neighbour outside the 5 x 5 grid.
5. Reject any neighbour that is an obstacle.
6. For each valid neighbour, calculate its g(n), h(n) and f(n).
7. Insert the valid nodes into the priority queue. If a node is reached with a lower cost, update it.
8. Continue the process until G is selected for expansion.
9. Trace parent links from G back to S to obtain the optimal path.

2.4 Priority Queue Update Format

Node	g(n)	h(n)	f(n) = g(n) + h(n)	Action
S	0	Distance from S to G	0 + h(S)	Expand S
Valid neighbour A	1	Distance from A to G	1 + h(A)	Insert
Valid neighbour B	1	Distance from B to G	1 + h(B)	Insert
Valid neighbour C	2	Distance from C to G	2 + h(C)	Insert / update if better

2.5 Exact Path and Expansion Order

The exact node expansion sequence, priority-queue contents, optimal path and final cost depend on the locations of S, G and the obstacles in the specific 5 x 5 grid. The grid layout is not included in the text supplied here, so giving a specific path would require inventing obstacle positions.

Assessment-safe statement: Once the given 5 x 5 grid is available, apply the above calculations to every expanded node and continue until G is reached. Because every move has unit cost and Manhattan distance is admissible for four-direction movement, A* returns an optimal path.

Question 3:Autonomous Rescue Robot - Online Search in a Disaster Building

Problem: An autonomous rescue robot must travel from Start (S) to a survivor at Goal (G) in a partially observable building. Obstacles block some cells, the robot can sense only adjacent cells, normal moves cost 1, risky zones have an additional cost of 2, and the robot must update its knowledge dynamically.

3.1 Analysis of Constraints

Constraint	Effect on Decision Making
Up, Down, Left, Right only	The robot cannot move diagonally; fewer actions are available at each state.
Obstacles	Blocked cells cannot be entered, so the robot must choose an alternative route.
Limited sensing	The robot knows only local information and must discover the environment while moving.
Normal move cost = 1	Shorter paths are generally preferred when they are safe.
Risky zone adds 2	Entering a risky zone has total movement cost 1 + 2 = 3.
Minimize total cost	The robot should compare cumulative costs, not only the number of steps.
Dynamic knowledge	The robot must update its internal map and re-plan when new information is observed.

3.2 Suitable Search Approach

Approach: Uniform Cost Search combined with an Online Search Agent.

Uniform Cost Search is appropriate because actions can have different costs: a normal move costs 1, while entering a risky zone costs 3 in total. The online component is necessary because the robot does not initially know the complete map.

3.3 Step-by-Step Working

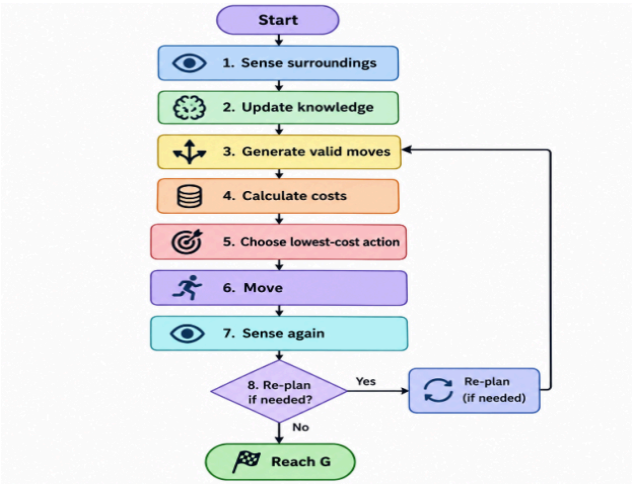
1. Start at S with cumulative cost g(S) = 0.
2. Sense the adjacent cells and identify open cells, obstacles and risky cells.
3. Generate only the actions that are currently possible.
4. Assign cost 1 to a normal move and cost 3 to a move into a risky zone.
5. Store candidate states in a priority queue ordered by the smallest cumulative cost g(n).
6. Expand the lowest-cost state and repeat the sensing and expansion process.
7. When new information reveals an obstacle or risk, update the internal map.
8. Re-plan from the current position using the updated knowledge.
9. Stop when the robot reaches G with the minimum safe total cost available from the discovered environment.

3.4 Simple Cost Example

Path Type	Calculation	Total Cost
3 normal moves	1 + 1 + 1	3
1 normal + 1 risky + 1 normal	1 + 3 + 1	5
2 risky moves	3 + 3	6

This shows why the robot must consider total cost and not just the number of moves.

3.5 Online Search Flow



3.6 AI Concepts Used

AI Concept	Application in the Rescue Robot
Intelligent Agent	The robot perceives its surroundings, makes decisions and performs actions.
Rationality	It selects the safest low-cost action that best helps it reach the survivor.
Perception	Sensors provide information about adjacent cells.
Planning/Search	The robot searches for a path from S to G.
Online learning/search	The robot updates its map as new information is discovered.

3.7 Environment Type

Property	Type	Reason
Observability	Partially observable	Only adjacent cells can be sensed.
Determinism	Possibly stochastic	Disaster conditions and newly revealed hazards can be uncertain.
Episodes	Sequential	Each move affects the next available states.
Change	Dynamic	The known situation can change as new information is discovered.
State	Discrete	The building is represented as individual grid cells.
Agents	Single-agent	The rescue robot is the main decision maker.
Knowledge	Initially incomplete	The robot discovers the map online.

3.8 Why This Approach Is Suitable

- Uniform Cost Search handles different action costs correctly.
- Online search handles partial information and local sensing.
- Dynamic replanning allows the robot to react when obstacles or risky zones are discovered.
- The approach supports rational, safe and cost-aware navigation toward the survivor.

Final Conclusion:

Uniform Cost Search with an online-search mechanism is suitable because it can compare different movement costs and re-plan when new information becomes available.